INSTITUTO SUPERIOR TECNOLÓGICO QUITO

Excelencia en Educación Superior

4.1 Variables Cuantitativas

python™

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In [1]:
```
# Creación aleatoria de datos
import numpy as np
X = np.array([0.5, 23, 0.3, 4.5, 0.3, 0.5,
              -28, -50, 60, -100, -10, -11, 13, 19, 1, 9])
```

In [2]:
```
# Medidas de tendencia central
from scipy import stats
media = np.mean(X)
mediana = np.median(X)
moda, _ = stats.mode(X)
print('media: ', media)
print('mediana: ', mediana)
print('moda :', moda)
```

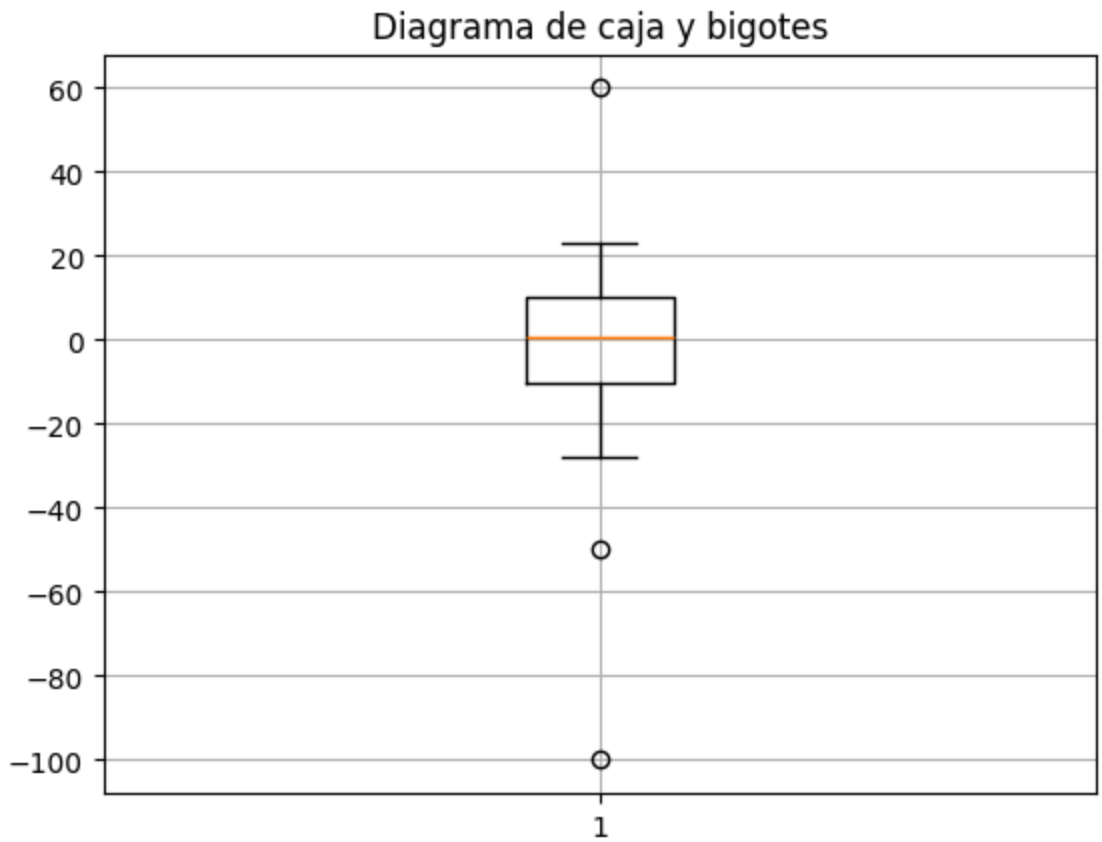
```
media:  -4.24375
mediana:  0.5
moda : 0.3
```

In [3]:
```
# Medidas de posición
import numpy as np
import matplotlib.pyplot as plt

Q1 = np.percentile(X, 25)
Q3 = np.percentile(X, 75)
RIC = Q3 - Q1
lim_inf = Q1 - 1.5*RIC
```

```
lim_sup = Q3 + 1.5*RIC

print('RIC :', RIC)
print('limite inferior :', lim_inf)
print('limite superior :', lim_sup)
```

```
RIC : 20.25
limite inferior : -40.625
limite superior : 40.375
```

In [4]:
```
# Dibuja el diagrama de box and whiskers y determina visualmente el número de outli
plt.boxplot(X)
plt.grid()
plt.title("Diagrama de caja y bigotes")
plt.show()
```



In [5]:
```
# Medidas de dispersión
desvest = np.std(X)
print('Desviación típica: ', desvest)
print('Varianza: ', desvest**2)
```

```
Desviación típica:  33.521242681880096
Varianza:  1123.6737109375
```

In [6]:
```
# Medidas de distribución
asimetria = stats.skew(X)
curtosis = stats.kurtosis(X, fisher=True) # https://docs.scipy.org/doc/scipy/refere
print('Asimetría ', np.round(asimetria, 4))
print('Curtosis ', np.round(curtosis, 4))
```

```python
if curtosis>0:
    print('Distribución Leptocúrtica')
elif curtosis<0:
    print('Distribución Platicúrtica')
else:
    print('Distribución Mesocúrtica')
```

```
Asimetría  -1.1302
Curtosis  2.2858
Distribución Leptocúrtica
```

In [7]:

```python
# Cargar el set de datos de Iris
from sklearn import datasets
from pprint import pprint
iris = datasets.load_iris()
pprint(iris)
X = iris.data
y = iris.target

# ¿Qué tipo de asimetría se observa en la variable "longitud del pétalo"?
#???

# ¿A qué tipo de distribución se ajusta la variable "ancho del sépalo", según su cu
#???
```

{'DESCR': '.. _iris_dataset:\n'
'\n'
'Iris plants dataset\n'
'--------------------\n'
'\n'
'**Data Set Characteristics:**\n'
'\n'
':Number of Instances: 150 (50 in each of three classes)\n'
':Number of Attributes: 4 numeric, predictive attributes and the '
'class\n'
':Attribute Information:\n'
'   - sepal length in cm\n'
'   - sepal width in cm\n'
'   - petal length in cm\n'
'   - petal width in cm\n'
'   - class:\n'
'   - Iris-Setosa\n'
'   - Iris-Versicolour\n'
'   - Iris-Virginica\n'
'\n'
':Summary Statistics:\n'
'\n'
'============== ==== ==== ======= ===== ====================\n'
'   Min Max   Mean   SD   Class Correlation\n'
'============== ==== ==== ======= ===== ====================\n'
'sepal length: 4.3 7.9   5.84   0.83   0.7826\n'
'sepal width: 2.0 4.4   3.05   0.43   -0.4194\n'
'petal length: 1.0 6.9   3.76   1.76   0.9490 (high!)\n'
'petal width: 0.1 2.5   1.20   0.76   0.9565 (high!)\n'
'============== ==== ==== ======= ===== ====================\n'
'\n'
':Missing Attribute Values: None\n'
':Class Distribution: 33.3% for each of 3 classes.\n'
':Creator: R.A. Fisher\n'
':Donor: Michael Marshall (MARSHALL%PLU@io.arc.nasa.gov)\n'
':Date: July, 1988\n'
'\n'
'The famous Iris database, first used by Sir R.A. Fisher. The '
'dataset is taken\n'
"from Fisher's paper. Note that it's the same as in R, but not as in "
'the UCI\n'
'Machine Learning Repository, which has two wrong data points.\n'
'\n'
'This is perhaps the best known database to be found in the\n'
"pattern recognition literature. Fisher's paper is a classic in the "
'field and\n'
'is referenced frequently to this day. (See Duda & Hart, for '
'example.) The\n'
'data set contains 3 classes of 50 instances each, where each class '
'refers to a\n'
'type of iris plant. One class is linearly separable from the other '
'2; the\n'
'latter are NOT linearly separable from each other.\n'
'\n'
'.. dropdown:: References\n'
'\n'

' - Fisher, R.A. "The use of multiple measurements in taxonomic '
'problems"\n'
'   Annual Eugenics, 7, Part II, 179-188 (1936); also in '
'"Contributions to\n'
'   Mathematical Statistics" (John Wiley, NY, 1950).\n'
' - Duda, R.O., & Hart, P.E. (1973) Pattern Classification and '
'Scene Analysis.\n'
'   (Q327.D83) John Wiley & Sons. ISBN 0-471-22361-1. See page '
'218.\n'
' - Dasarathy, B.V. (1980) "Nosing Around the Neighborhood: A New '
'System\n'
'   Structure and Classification Rule for Recognition in Partially '
'Exposed\n'
'   Environments". IEEE Transactions on Pattern Analysis and '
'Machine\n'
'   Intelligence, Vol. PAMI-2, No. 1, 67-71.\n'
' - Gates, G.W. (1972) "The Reduced Nearest Neighbor Rule". IEEE '
'Transactions\n'
'   on Information Theory, May 1972, 431-433.\n'
' - See also: 1988 MLC Proceedings, 54-64. Cheeseman et al"s '
'AUTOCLASS II\n'
'   conceptual clustering system finds 3 classes in the data.\n'
' - Many, many more ...\n',
'data': array([[5.1, 3.5, 1.4, 0.2],
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```

```
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[6.7, 3.1, 5.6, 2.4],
[6.9, 3.1, 5.1, 2.3],
[5.8, 2.7, 5.1, 1.9],
[6.8, 3.2, 5.9, 2.3],
[6.7, 3.3, 5.7, 2.5],
```

```
       [6.7, 3. , 5.2, 2.3],
       [6.3, 2.5, 5. , 1.9],
       [6.5, 3. , 5.2, 2. ],
       [6.2, 3.4, 5.4, 2.3],
       [5.9, 3. , 5.1, 1.8]]),
 'data_module': 'sklearn.datasets.data',
 'feature_names': ['sepal length (cm)',
                   'sepal width (cm)',
                   'petal length (cm)',
                   'petal width (cm)'],
 'filename': 'iris.csv',
 'frame': None,
 'target': array([0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
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       2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2]),
 'target_names': array(['setosa', 'versicolor', 'virginica'], dtype='<U10')}
```

In [8]:

```python
# ¿Qué tipo de asimetría se observa en la variable "longitud del pétalo"?
#???
from scipy import stats

# 1. Seleccionamos la columna de longitud del pétalo (índice 2)
petal_length = X[:, 2]

# 2. Calculamos la asimetría
asimetria_petal = stats.skew(petal_length)

print(f'Asimetría de la longitud del pétalo: {asimetria_petal:.4f}')

# Respuesta a la pregunta:
if asimetria_petal < 0:
    print("Se observa una ASIMETRÍA NEGATIVA (sesgada a la izquierda).")
elif asimetria_petal > 0:
    print("Se observa una ASIMETRÍA POSITIVA (sesgada a la derecha).")
else:
    print("La distribución es simétrica.")
```

```
Asimetría de la longitud del pétalo: -0.2721
Se observa una ASIMETRÍA NEGATIVA (sesgada a la izquierda).
```

In [9]:

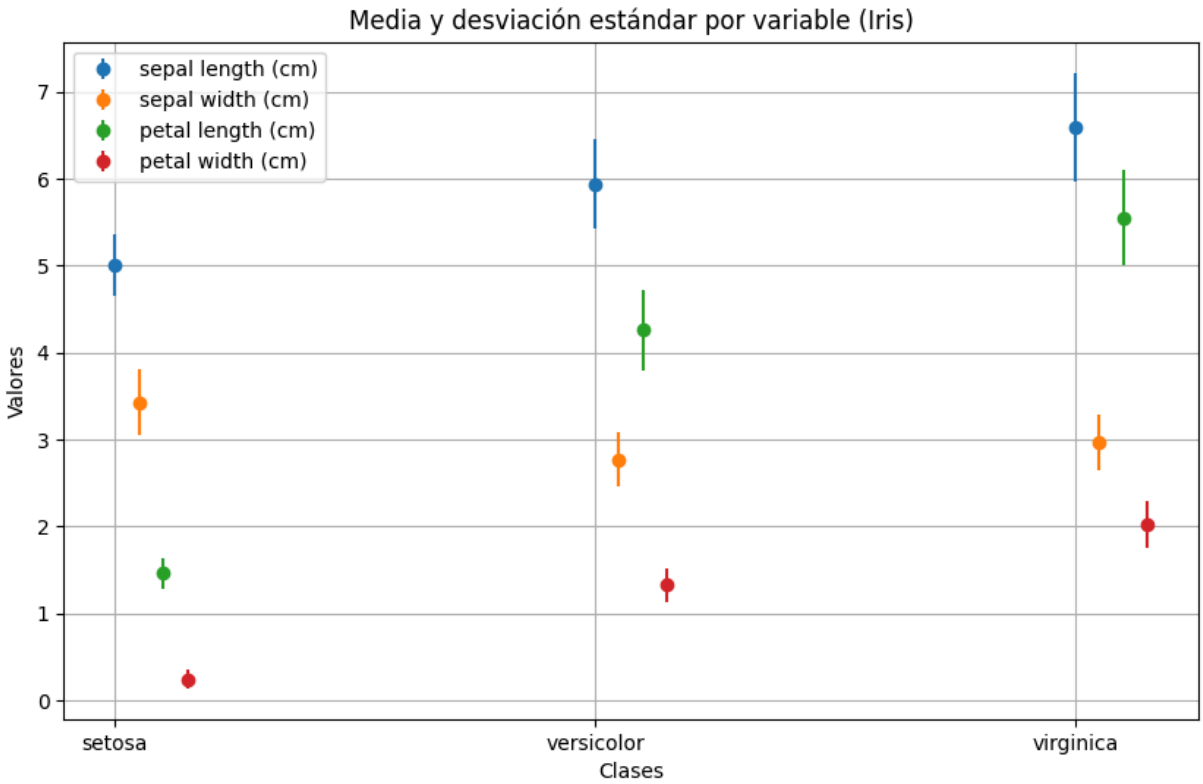
```python
# ¿A qué tipo de distribución se ajusta la variable "ancho del sépalo", según su cu
#???
from scipy import stats

# 1. Seleccionamos la columna de ancho del sépalo (índice 1)
sepal_width = X[:, 1]

# 2. Calculamos la curtosis (Fisher por defecto, normal = 0)
curtosis_sepal = stats.kurtosis(sepal_width)

print(f'Curtosis del ancho del sépalo: {curtosis_sepal:.4f}')
```

```python
# 3. Clasificación
if curtosis_sepal > 0:
    print("La distribución es LEPTOCÚRTICA (más apuntada que la normal).")
elif curtosis_sepal < 0:
    print("La distribución es PLATICÚRTICA (más achatada que la normal).")
else:
    print("La distribución es MESOCÚRTICA (similar a la normal).")
```

```
Curtosis del ancho del sépalo: 0.1810
La distribución es LEPTOCÚRTICA (más apuntada que la normal).
```

In [12]:

```python
# EJERCICIO EXTRA SOBRE EL DATASET DE IRIS
# Extraer la media y la desviación típica de cada variable en función de la clase y
import numpy as np
import matplotlib.pyplot as plt
from sklearn import datasets

# Cargar dataset iris
iris = datasets.load_iris()
X = iris.data
y = iris.target
feature_names = iris.feature_names
clases = np.unique(y)

# Crear figura
plt.figure(figsize=(10,6))

# Recorrer cada variable (columna)
for i in range(X.shape[1]):
    medias = []
    desv = []

    # Calcular media y desviación por clase
    for c in clases:
        datos_clase = X[y == c, i]
        medias.append(np.mean(datos_clase))
        desv.append(np.std(datos_clase))

    # Graficar con errorbar
    plt.errorbar(clases + i*0.05, medias, yerr=desv, fmt='o', label=feature_names[i

# Configuración del gráfico
plt.xticks(clases, iris.target_names)
plt.xlabel("Clases")
plt.ylabel("Valores")
plt.title("Media y desviación estándar por variable (Iris)")
plt.legend()
plt.grid()

plt.show()
```



5. Github

https://github.com/Erick-305/machine-learning.git